

E-Commerce Revenue Prediction Using Machine Learning

1. Introduction

Online shopping websites generate a large amount of customer browsing data every day. Businesses can use this data to understand customer behavior. It also predicts whether a visitor generates revenue or not. Accurate prediction of customer purchases helps companies to improve marketing, sales, and customer experiences.

In this project, an e-commerce dataset is used to develop a machine learning model that predicts whether a website visitor will generate revenue. The dataset contains information about visitor activity, browsing behavior, page visits, and other related attributes. Different data preprocessing and machine learning techniques are applied to build and evaluate the prediction model.

2. Problem Statement

E-commerce businesses face difficulty in identifying which visitors are likely to make a purchase. Predicting customer purchase behavior can help organizations to target potential customers. Therefore, this project aims to predict whether revenue will be generated based on visitor browsing data.

3. Objectives

- To analyze the e-commerce dataset.
- To perform data preprocessing and prepare the data for machine learning.
- To identify important features affecting revenue generation.
- To train and evaluate a machine learning model.
- To predict whether a visitor will generate revenue or not.
- To assess the performance of the developed model using evaluation metrics.

4. Custom Features Created

4.1 Feature Engineering

4.1.1 Total Time Feature Creation

Description:

A new feature named Total_Time was created by combining Administrative_Duration, Informational_Duration, and ProductRelated_Duration. This feature represents the total time spent by a visitor on the website.

Why It Was Created:

Individual duration features only show time spent on specific page categories. Combining them provides an overall measure of user engagement.

Role in Model Training:

Visitors who spend more time on the website are generally more interested in products and are more likely to generate revenue. Therefore, this feature helps the model identify highly engaged customers.

4.1.2 Total Pages Feature Creation**Description:**

A new feature named Total_Pages was created by adding Administrative, Informational, and ProductRelated page visits.

Why It Was Created:

This feature measures the total number of pages visited by a customer during a session.

Role in Model Training:

Customers who visit more pages usually show higher interest in the website content. This feature helps the model understand browsing intensity and customer engagement.

4.1.3 Engagement Ratio Feature Creation**Description:**

The Engagement_Ratio feature was created using:

$$\text{Engagement Ratio} = \frac{\text{Total Time}}{\text{Total Pages} + 1}$$

Why It Was Created:

This feature measures the average time spent per page.

Role in Model Training:

A higher engagement ratio indicates that users spend more time reading or exploring each page, which may increase the likelihood of a purchase.

4.1.4 Value Per Page Feature Creation**Description:**

The Value_Per_Page feature was created by dividing PageValues by ProductRelated pages.

Why It Was Created:

This feature estimates the value generated by each product-related page visit.

Role in Model Training:

It helps the model understand whether product page visits contribute to revenue generation and customer purchase intent.

4.2 Polynomial Feature Generation**Description:**

Polynomial Feature Generation technique was applied on the engineered features Total_Time, Total_Pages, Engagement_Ratio, and Value_Per_Page. Degree 2 polynomial features were created to capture interactions between variables.

Why It Was Created:

Some relationships in customer behavior are not purely linear. Polynomial features help represent hidden interactions between variables that may not be visible in the original dataset.

Role in Model Training:

These features provide additional information to machine learning algorithms and help improve prediction performance by capturing complex customer behavior patterns.

4.2.1 Total_Time × Total_Pages (Total_Time_Total_Pages)**Description:**

This interaction feature combines the total time spent on the website with the total number of pages visited.

Why It Was Created:

Customers who spend more time and visit more pages are usually more engaged.

Role in Model Training:

It helps the model identify highly active users who are more likely to generate revenue.

4.2.2 Total_Time × Engagement_Ratio (Total_Time_Engagement_Ratio)**Description:**

This feature represents the interaction between total browsing time and user engagement level.

Why It Was Created:

A customer may spend a long time on the website but still have low engagement. This feature captures both factors together.

Role in Model Training:

It helps distinguish genuinely interested customers from casual visitors.

4.2.3 Total_Pages × Engagement_Ratio (Total_Pages_Engagement_Ratio)**Description:**

This interaction feature combines the number of pages visited with the engagement ratio.

Why It Was Created:

It provides a more detailed measure of browsing behavior.

Role in Model Training:

The model can better understand how page visits and engagement jointly influence purchasing decisions.

4.2.4 Value_Per_Page × Engagement_Ratio (Value_Per_Page_Engagement_Ratio)**Description:**

This feature combines page value information with user engagement.

Why It Was Created:

Customers with high-value page interactions and strong engagement are more likely to complete a purchase.

Role in Model Training:

It improves the model's ability to identify sessions with high revenue potential.

4.2.5 Squared Features (Degree 2)**Description:**

Polynomial transformation also created squared versions of the selected features such as Total_Time², Total_Pages², Engagement_Ratio², and Value_Per_Page².

Why It Was Created:

Squared features capture non-linear relationships that cannot be represented by the original variables alone.

Role in Model Training:

They help machine learning models learn complex patterns and improve predictive accuracy.

Note:

Polynomial features were automatically generated using the PolynomialFeatures() function. Only the most relevant polynomial features were retained during feature selection to avoid overfitting and redundancy.

4.3 Feature Binning**4.3.1 ProductRelated_Duration_Bin****Description:**

ProductRelated_Duration was converted into three categories: Low, Medium, and High.

Why It Was Created:

Binning simplifies continuous numerical values into meaningful groups.

Role in Model Training:

It helps the model identify different levels of customer engagement based on product

browsing duration.

4.3.2 BounceRates_Bin

Description:

BounceRates were categorized into Low, Medium, and High groups.

Why It Was Created:

Bounce rate is an important indicator of customer interest and website engagement.

Role in Model Training:

Visitors with lower bounce rates are generally more engaged and have a higher probability of generating revenue. This feature helps the model distinguish between engaged and non-engaged users.

5. Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning. Several techniques were applied to clean, transform, and optimize the e-commerce dataset before model training.

5.1 Missing Value Analysis

Missing values can reduce the quality of data and negatively affect machine learning algorithms. Therefore, the dataset was examined to identify any null or missing values.

The analysis confirmed that the dataset was complete and suitable for further processing. This ensured that model training was performed on reliable data without information loss.

5.2 Outlier Analysis

Outliers are extreme observations that differ significantly from normal data patterns. They can distort statistical analysis and affect model performance.

Outlier detection was performed using the Interquartile Range (IQR) method and boxplots. This helped identify unusual values present in numerical attributes and provided a better understanding of data distribution.

5.3 Log Transformation

Several numerical features were positively skewed and contained large value ranges. Skewed data may reduce the effectiveness of machine learning algorithms.

Log transformation was applied to reduce skewness and make feature distributions more balanced. This improved data consistency and reduced the influence of extreme values.

5.4 Outlier Capping

Even after transformation, some extreme values remained in the dataset. Completely removing records could lead to loss of useful information.

Outliers were capped using IQR-based lower and upper bounds. This reduced the effect of extreme observations while preserving all records in the dataset.

5.5 Feature Engineering

Raw features do not always provide sufficient information for machine learning models. Creating new features can help capture hidden patterns in customer behavior.

Additional features such as Total Time, Total Pages, Engagement Ratio, and Value Per Page were generated. These features provided more meaningful information about user engagement and purchasing behavior.

5.6 Encoding Categorical Data

Machine learning algorithms require numerical input and cannot directly process categorical values.

Different encoding techniques were applied to convert categorical attributes into numerical form. Label Encoding was used for binary and ordinal variables, while One-Hot Encoding was applied to nominal variables. This allowed the model to effectively utilize categorical information.

5.7 Feature Scaling

Numerical features existed on different scales. Large-scale variables could dominate smaller-scale variables during model training.

Standard Scaling was applied to normalize numerical features. This ensured that all variables contributed fairly to the learning process and improved model performance.

5.8 Correlation Analysis

Correlation analysis helps identify relationships between variables and detect highly correlated features.

A correlation heatmap was generated to analyze feature relationships. This provided useful insights into the dataset and supported feature selection decisions.

5.9 Variance Threshold Feature Selection

Features with very low variance contain little useful information and may not contribute to prediction performance.

Low-variance features were removed, reducing dataset complexity and improving computational efficiency.

5.10 Correlation-Based Feature Filtering

Highly correlated features often contain redundant information and may increase model complexity.

Features with correlation greater than the defined threshold were removed. This reduced redundancy and improved model interpretability.

5.11 Univariate Feature Selection

Not all features contribute equally to predicting revenue. Selecting important features helps improve model performance.

Mutual Information-based feature selection was used to retain the most informative attributes, reducing noise and improving prediction quality.

5.12 Recursive Feature Elimination (RFE)

Feature selection helps identify the most relevant variables and remove less useful ones.

RFE selected the most important features for classification, resulting in a simpler and more efficient model.

5.13 Train-Test Split

A separate testing dataset is required to evaluate model performance on unseen data.

The dataset was divided into training and testing sets using an 80:20 ratio. This enabled a fair assessment of the model's generalization ability.

5.14 Data Balancing (SMOTE)

The Revenue target variable was imbalanced, meaning one class contained significantly more records than the other. This can cause biased predictions.

SMOTE was applied to generate synthetic samples for the minority class and balance the dataset. This improved the model's ability to correctly identify both revenue classes and increased recall performance.

6. Exploratory Data Analysis (EDA)

6.1 Revenue Distribution

Interpretation:

The Revenue distribution graph shows that the majority of website visitors did not generate revenue. While only a smaller number of visitors completed a purchase. This indicates that the target variable is imbalanced.

Real-World Reasoning:

In e-commerce websites, many users browse products without making a purchase. Only a small percentage of visitors convert into customers.

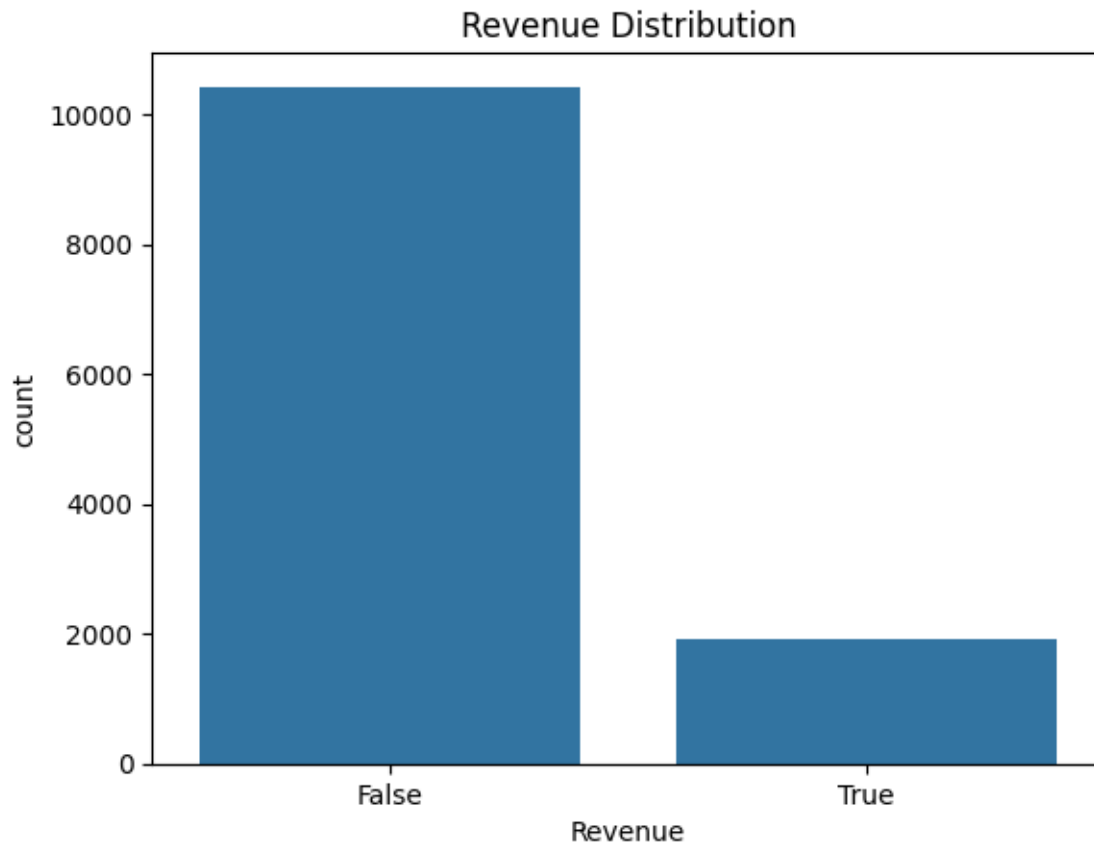


Figure 1: Revenue Distribution

6.2 Feature Distribution Analysis (Histograms)

Interpretation:

Histograms were created for features such as Administrative, Administrative_Duration, ProductRelated, ProductRelated_Duration, BounceRates, ExitRates, PageValues, and SpecialDay. Most variables show right-skewed distributions where a large number of observations are concentrated at lower values.

Real-World Reasoning:

Most visitors spend limited time on the website, while only a few users explore many pages and spend significant time before making a purchase decision.

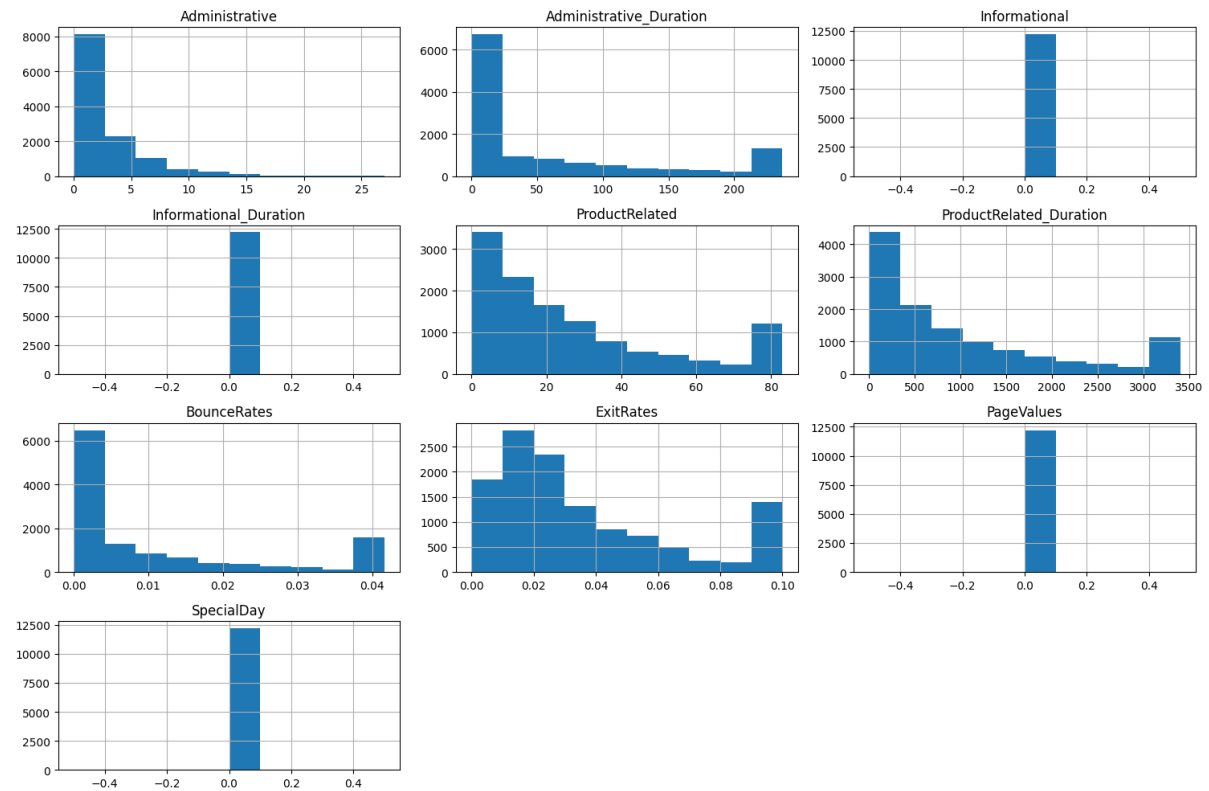


Figure 2: Feature Distribution Analysis

6.3 Outlier Analysis (Box Plots)

Interpretation:

Box plots revealed the presence of multiple outliers in features such as Administrative_Duration, ProductRelated_Duration, BounceRates, ExitRates, and PageValues. The calculated skewness values also confirmed positive skewness in most numerical features.

Real-World Reasoning:

Some customers spend exceptionally long periods browsing products or visiting many pages. These behaviors create extreme values that appear as outliers.

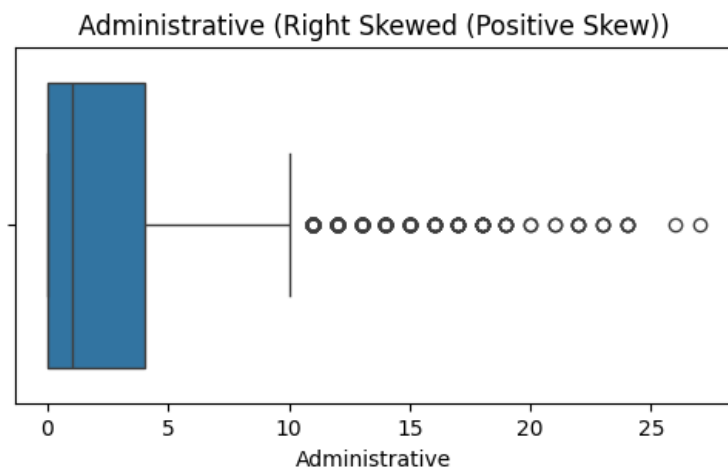


Figure 3: Administrative outlier detection

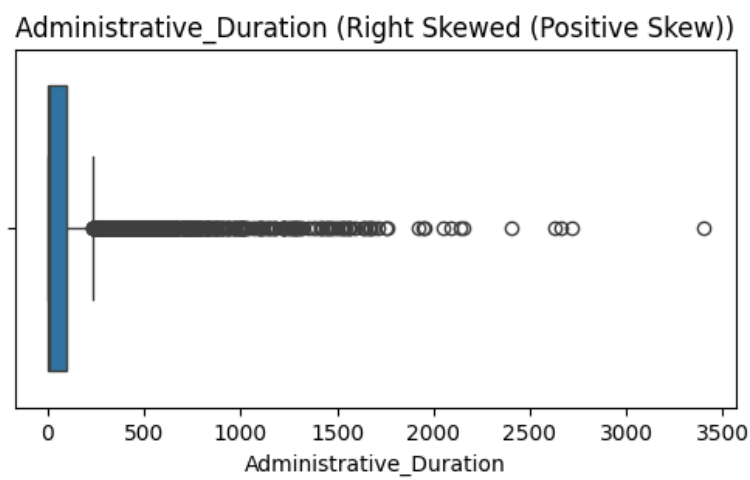


Figure 4: Administrative_Duration Outlier detection

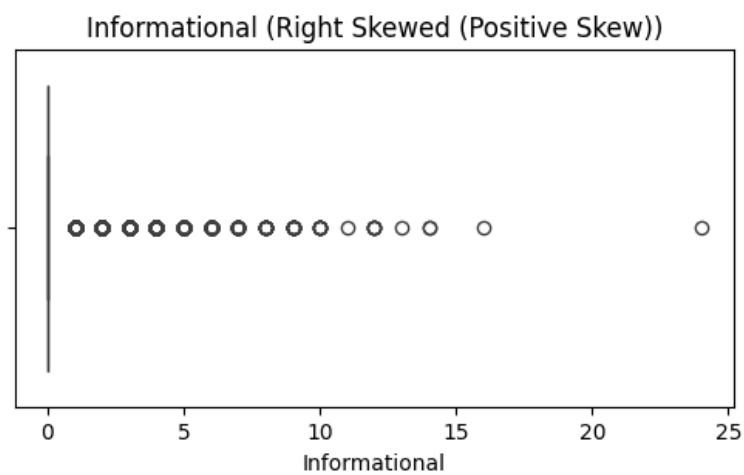


Figure 5: Informational outlier detection

Informational_Duration (Right Skewed (Positive Skew))

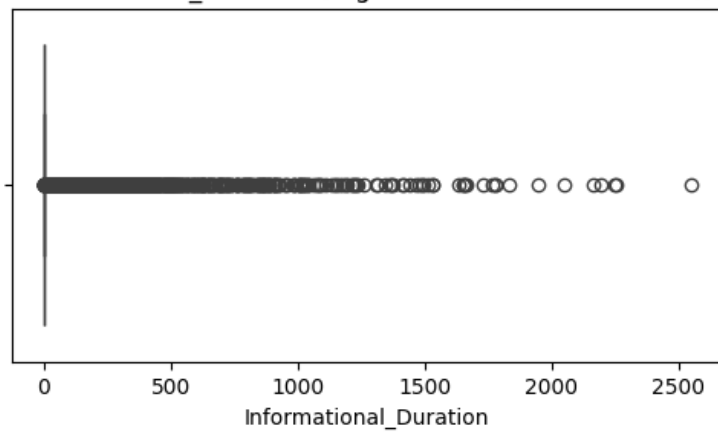


Figure 6: Information_Duration outlier detection

ProductRelated (Right Skewed (Positive Skew))

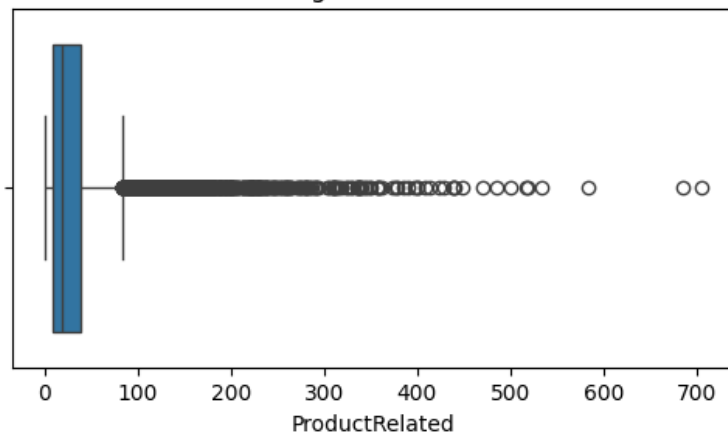


Figure 7: Product Related outlier detection

ProductRelated_Duration (Right Skewed (Positive Skew))

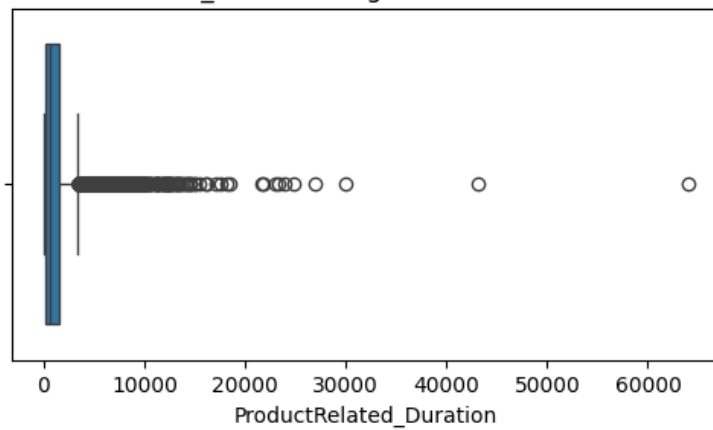


Figure 8: ProductRelated_Duration outlier detection

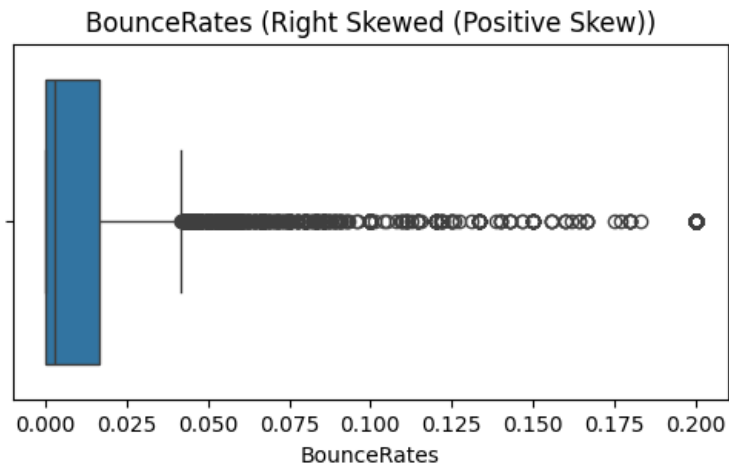


Figure 9: BounceRates outlier detection

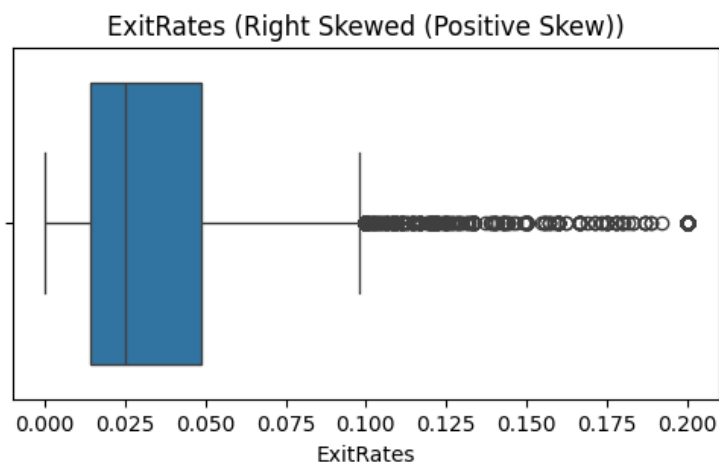


Figure 10: ExitRates outlier detection

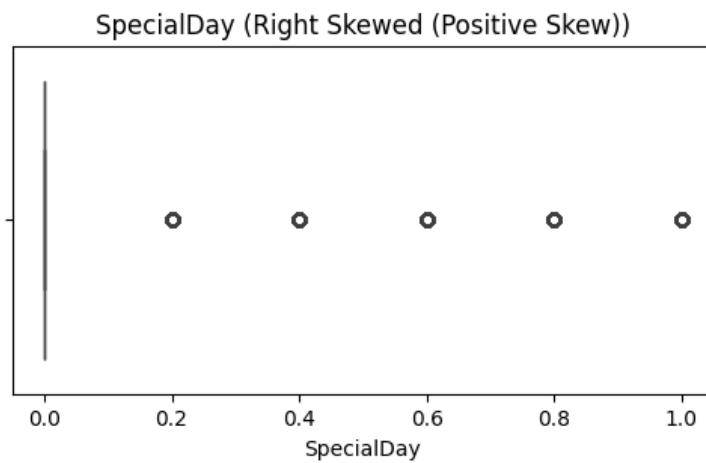


Figure 11: SpecialDay outlier detection

6.4 Correlation Heatmap

Interpretation:

The heatmap shows strong positive correlations between:

- ProductRelated and ProductRelated_Duration (0.89)
- Administrative and Administrative_Duration (0.82)

A strong positive relationship was also observed between BounceRates and ExitRates (0.74).

Real-World Reasoning:

Visitors who view more product pages naturally spend more time on those pages. Similarly, users with high bounce rates often have higher exit rates.

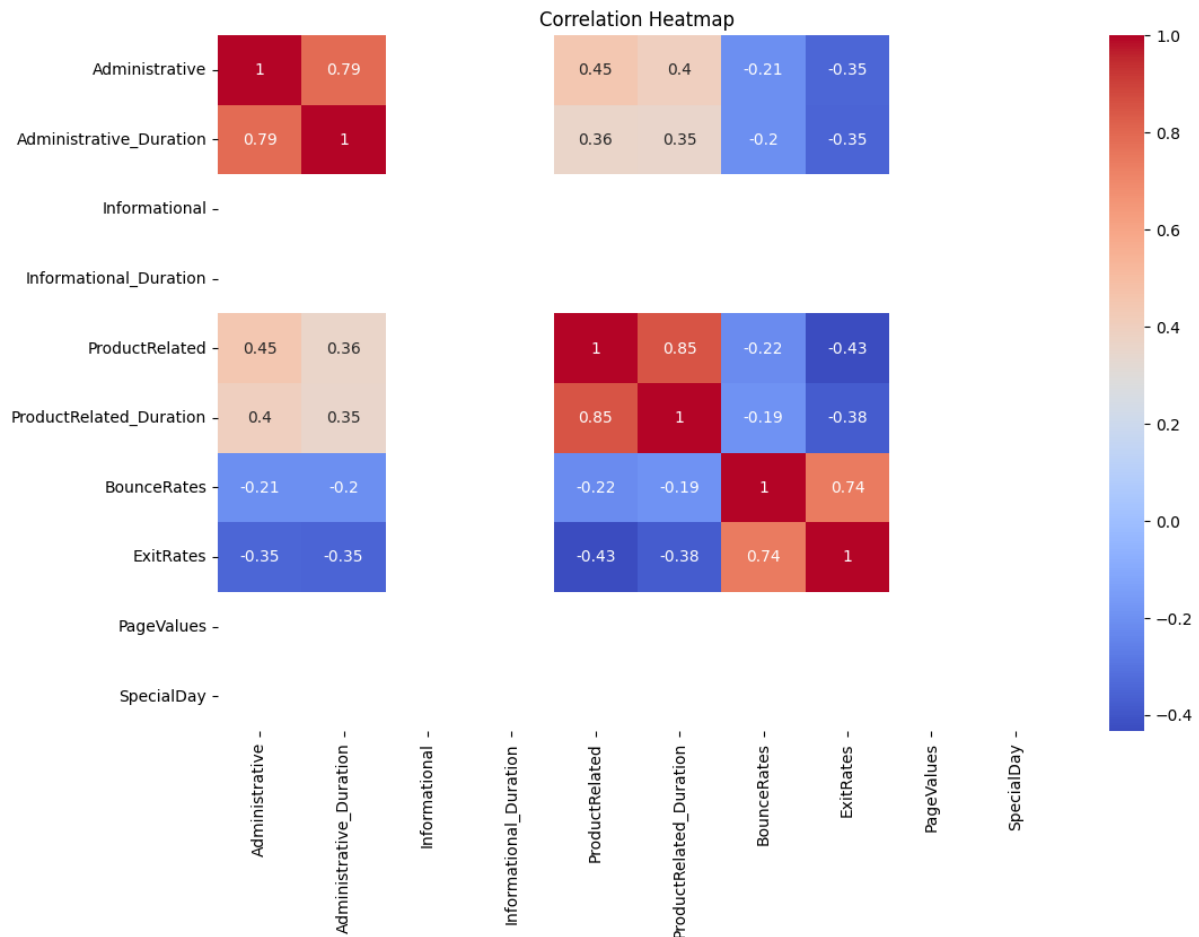


Figure 12: Correlation Heatmap

6.5 Visitor Type vs Revenue

Interpretation:

Returning visitors generated significantly more revenue compared to new visitors and other visitor types.

Real-World Reasoning:

Returning visitors are already familiar with the website and products, making them more likely to complete purchases.

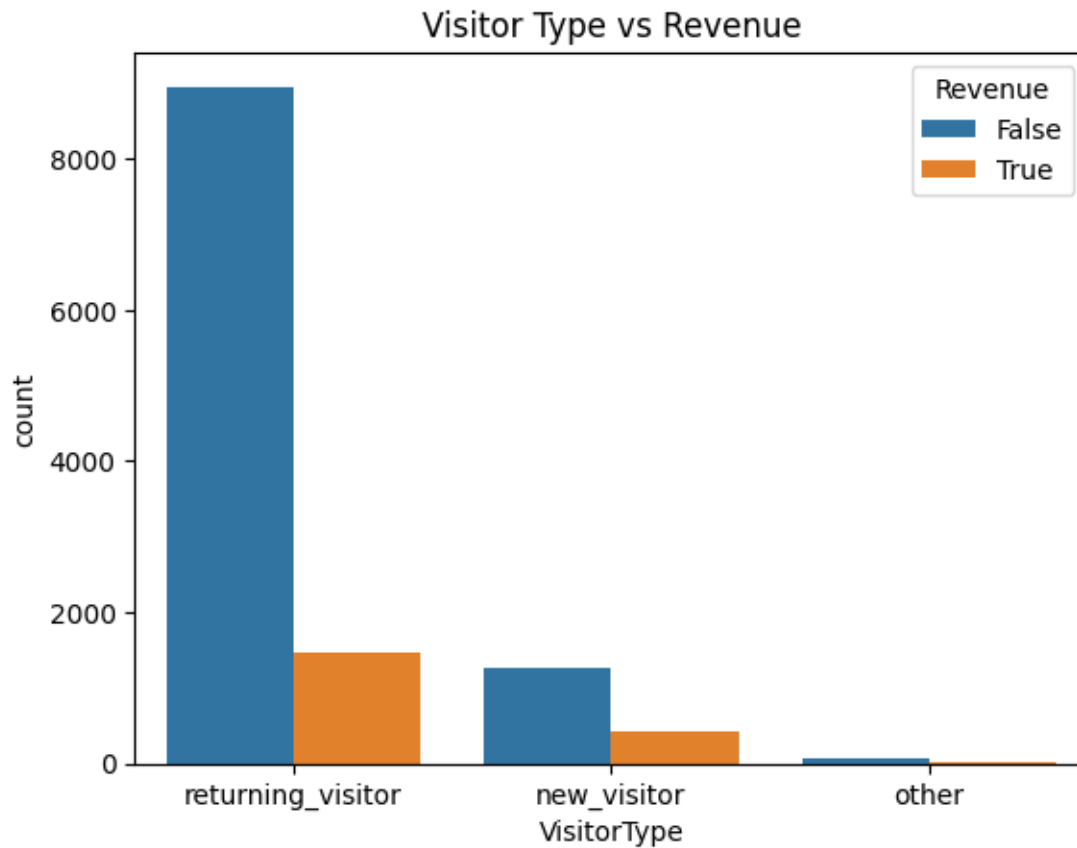


Figure 13: Visitor Type vs Revenue

6.6 Month vs Revenue

Interpretation:

Revenue generation varied across different months. Higher purchase activity was observed during months such as November and December.

Real-World Reasoning:

These months often include seasonal sales, holiday shopping periods, and promotional campaigns that encourage customer purchases.

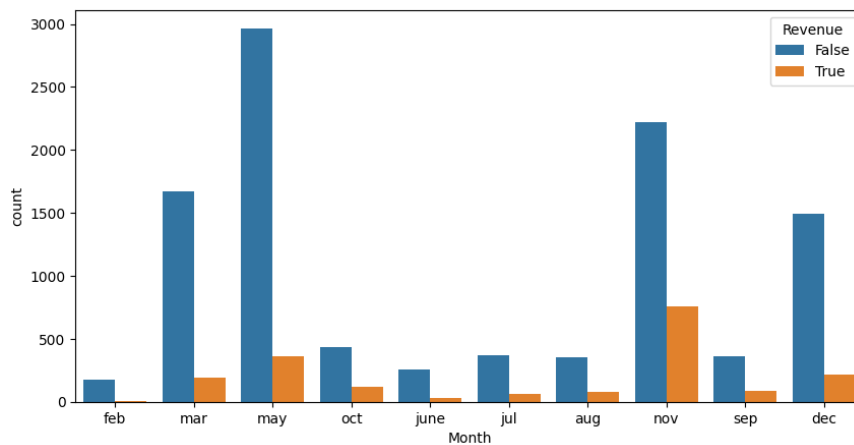


Figure 14: Month vs Revenue

6.7 Feature Relevance Using Mutual Information

Interpretation:

The most relevant features for revenue prediction were:

- ExitRates
- ProductRelated_Duration
- Total_Pages
- BounceRates_Bin
- ProductRelated

Real-World Reasoning:

Customer engagement and browsing behavior have a strong influence on purchasing decisions.

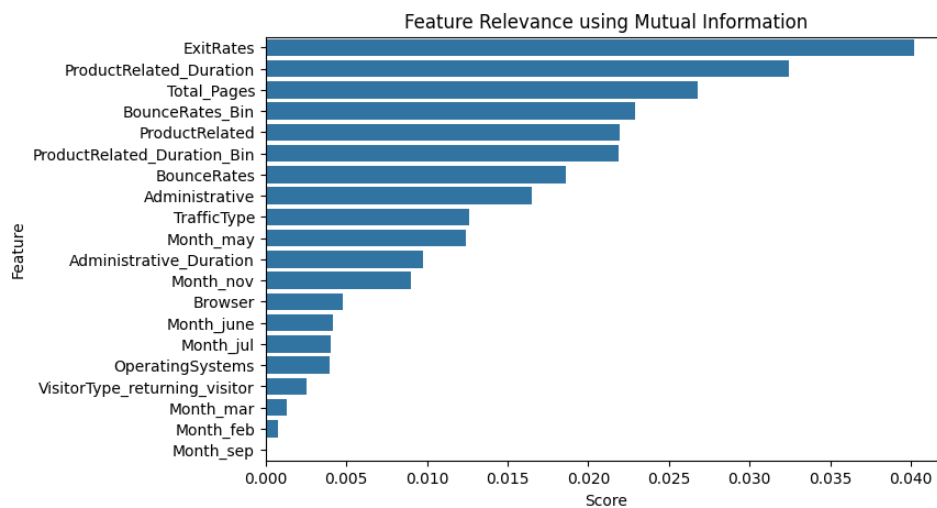


Figure 15: Feature Relevance Using Mutual Information

6.8 Top 10 Important Features

Interpretation:

Feature importance analysis identified ExitRates, ProductRelated_Duration, Total_Pages, and ProductRelated as the most influential predictors of revenue generation.

Real-World Reasoning:

Visitors who spend more time viewing products and interact with more pages are generally more likely to make a purchase.

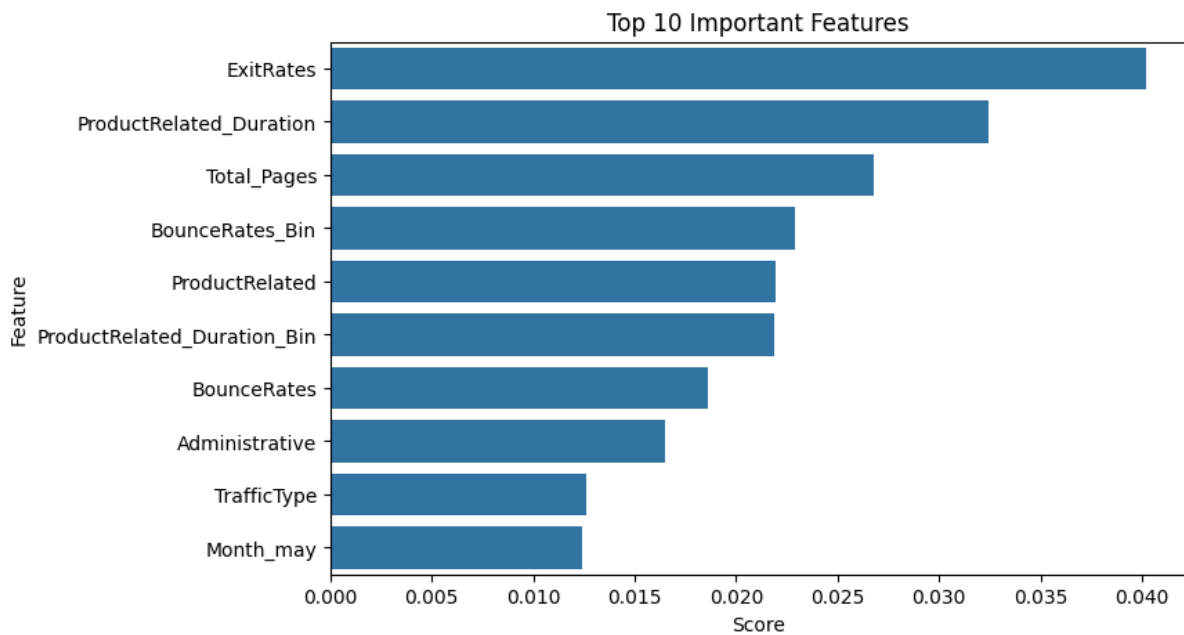


Figure 16: Top 10 Important Features

Custom Insights

Insight 1

Returning visitors are more likely to generate revenue than new visitors, indicating the importance of customer retention strategies.

Insight 2

Product-related browsing activity is strongly associated with revenue generation. Visitors who view more products and spend more time exploring them are more likely to purchase.

Insight 3

High Bounce Rates and Exit Rates negatively impact revenue generation, suggesting that improving website engagement could increase conversions.

Insight 4

Revenue generation increases during certain months, particularly November and December, highlighting the impact of seasonal shopping behavior.

Insight 5

The dataset is highly imbalanced, with significantly more non-purchasing visitors than purchasing visitors. This justified the use of SMOTE during preprocessing.

7. Machine Learning Task

7.1 Prediction Objective

The main objective of this project is to predict whether an e-commerce website visitor will generate revenue or not. The prediction is based on customer browsing behavior, page visits, session duration, visitor type, page values, bounce rates, exit rates, and other related features.

Since the target variable (Revenue) contains two classes (True and False). This project is treated as a binary classification problem.

7.2 Machine Learning Algorithms Applied

To identify the most suitable model for revenue prediction, multiple machine learning algorithms were trained and evaluated.

7.2.1 Logistic Regression

Logistic Regression was used as a baseline classification model. It estimates the probability of revenue generation based on input features. It provides a simple and interpretable approach for binary classification.

7.2.2 Gaussian Naive Bayes

Gaussian Naive Bayes is a probabilistic classification algorithm that assumes independence among features. It is computationally efficient and performs well on structured datasets. This model was included to compare its performance with other classification techniques.

7.2.3 Decision Tree Classifier

Decision Tree was used to learn decision rules from the dataset. It can capture non-linear relationships between features and the target variable while providing easy-to-understand decision paths.

7.2.4. Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees. It improves prediction accuracy, reduces overfitting, and provides feature importance information for identifying influential variables.

7.3 Cross Validation

To ensure reliable model evaluation, 5-Fold Cross Validation was applied using the `cross_val_score()` function.

The dataset was divided into five folds. During each iteration, four folds were used for training and one fold was used for validation. This process was repeated five times so that every fold was used as the validation set once.

Recall was selected as the evaluation metric because correctly identifying revenue-generating visitors was an important objective of this project.

A single train-test split may produce biased results depending on the selected samples. Cross-validation provides a more stable and reliable estimate of model performance by evaluating the model on multiple subsets of the dataset.

Impact

Cross-validation improved the reliability of model evaluation, reduced the risk of overfitting, and enabled a fair comparison of different machine learning algorithms.

7.4 Model Evaluation

The performance of all machine learning models was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These evaluation measures helped compare the effectiveness of different algorithms and identify the most suitable model for revenue prediction.

Model Performance Analysis

Initially, all four machine learning models were evaluated using Accuracy, Precision, Recall, and F1-Score.

Random Forest achieved the highest Accuracy (75.22%), while Gaussian Naive Bayes achieved the highest Recall (82.46%). Since the primary objective of this project was to correctly identify revenue-generating visitors, Recall was considered more important than Accuracy.

A high Recall value indicates that the model successfully identifies a larger proportion of actual revenue-generating customers. Missing potential customers can have a direct impact on business revenue; therefore, maximizing Recall was prioritized.

To further validate model performance, 5-Fold Cross Validation was applied. Gaussian Naive Bayes achieved the highest Cross Validation Recall score of 0.915, outperforming Logistic Regression (0.7537), Decision Tree (0.7974), and Random Forest (0.8917).

The consistency between the test Recall (0.8246) and Cross Validation Recall (0.915) indicates that Gaussian Naive Bayes generalized well across different subsets of the dataset.

Therefore, Gaussian Naive Bayes was selected as the final model because it achieved:

- Highest Recall (0.8246)
- Highest Cross Validation Recall (0.915)
- Consistent performance across folds
- Better ability to identify revenue-generating visitors

Although Random Forest achieved higher Accuracy, its Recall was significantly lower. Since identifying potential purchasing customers was the main objective of this project, Gaussian Naive Bayes was considered the most suitable model.

7.5. Results and Discussions

7.5.1 Model Performance Comparison

Four machine learning algorithms were evaluated for revenue prediction: Logistic Regression, Gaussian Naive Bayes, Decision Tree, and Random Forest.

Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.6313	0.6313	0.6313	0.6313
Gaussian Naive Bayes	0.5539	0.5539	0.5539	0.5539
Decision Tree	0.7206	0.7206	0.7206	0.7206
Random Forest	0.7522	0.7522	0.7522	0.7522

Cross Validation Results

Model	Mean Recall
Logistic Regression	0.7537
Decision Tree	0.7974
Random Forest	0.8917
Gaussian Naive Bayes	0.9150

7.5.2 Why Gaussian Naive Bayes Performed Better

Although Random Forest achieved the highest Accuracy (75.22%), Gaussian Naive Bayes achieved the highest Recall (82.46%) and the highest Cross Validation Recall (91.50%). The primary objective of this project was to correctly identify visitors who are likely to generate revenue. Therefore, Recall was considered more important than Accuracy.

A higher Recall means the model successfully identifies a larger number of actual revenue-generating customers. Missing potential customers can reduce business opportunities and affect revenue generation. Gaussian Naive Bayes also showed consistent performance during 5-Fold Cross Validation, indicating better generalization and model stability. Therefore, it was selected as the final model.

7.5.3 Feature Importance

Feature importance analysis revealed that the following features had the strongest influence on revenue prediction:

- ExitRates
- ProductRelated_Duration
- Total_Pages
- BounceRates_Bin
- ProductRelated
- Administrative
- TrafficType

7.5.4 Interpretation

These features are closely related to customer engagement. Visitors who spend more time viewing products, browse more pages, and interact extensively with the website are more likely to make purchases. Similarly, higher Exit Rates and Bounce Rates generally indicate a lower probability of revenue generation because users leave the website without completing transactions.

6. 5.4 Ethical Concerns

The model uses customer browsing behavior to predict purchasing activity. Although personal sensitive information is not included, ethical considerations remain important.

- Customer privacy should be protected.
- User data should be collected with proper consent.
- Predictions should not be used for unfair discrimination.
- Businesses should maintain transparency regarding how customer data is utilized.

Responsible use of customer information is essential for maintaining user trust.

7.5.5 Dataset Limitations

The dataset has several limitations:

- It represents data from a specific e-commerce environment.
- Customer behavior may change over time.
- External factors such as promotions, discounts, seasonal trends, and economic conditions are not included.
- Some customer characteristics that influence purchasing decisions may not be available in the dataset.

These limitations may affect model performance when applied to different situations.

7.5.6 Bias Issues

The original dataset was highly imbalanced.

- Non-revenue visitors: 84.53%
- Revenue-generating visitors: 15.47%

This imbalance could cause the model to favor the majority class. To address this issue, SMOTE was applied to balance the training data. Although SMOTE improved class representation, synthetic samples may not perfectly reflect real customer behavior. Therefore, some bias may still remain.

7.5.7 Generalization Capability

The model's ability to generalize was evaluated using 5-Fold Cross Validation. Gaussian Naive Bayes achieved the highest Cross Validation Recall score of 0.915 with consistent performance across different folds. This indicates that the model is capable of making reliable predictions on unseen data. However, customer preferences and online shopping trends change over time. Therefore, periodic retraining with updated data would be necessary to maintain prediction performance.

8. Advanced Component: Feature Selection

Feature selection was performed to identify the most relevant attributes for revenue prediction and remove redundant or less informative features from the dataset. This helped improve model efficiency, reduce dimensionality, and enhance prediction performance.

The original dataset contained 42 features. Not all features contribute equally to predicting revenue generation. Some features may contain redundant information, while others may have very little predictive value. Using unnecessary features can increase model complexity and may lead to overfitting.

Therefore, multiple feature selection techniques were applied to retain only the most important attributes.

8.1 Variance Threshold

A Variance Threshold of 0.01 was applied to remove features with very low variance.

Features with very little variation provide limited information for classification because their values remain nearly constant across observations.

The number of features was reduced from 42 to 32 by removing low-information attributes.

8.2 Correlation Filtering

A correlation matrix was generated and features having a correlation greater than 0.90 were removed.

Highly correlated features often contain similar information and can create redundancy in the dataset.

The feature count was further reduced from 32 to 26, helping eliminate duplicate information and improve model efficiency.

8.3 Univariate Feature Selection

Mutual Information was used through SelectKBest to identify features that had the strongest relationship with the target variable (Revenue).

Not all features contribute equally to prediction. Mutual Information helps measure the importance of each feature individually.

The top 20 most informative features were selected for further processing.

8.4 Recursive Feature Elimination (RFE)

Recursive Feature Elimination (RFE) was applied using Logistic Regression as the base estimator.

RFE iteratively removes less important features and retains only those that contribute most to model performance.

The final feature set was reduced to 10 highly relevant features that were used for model training and evaluation.

Final Selected Features

The final selected features included important customer behavior attributes such as:

- ProductRelated
- ProductRelated_Duration
- BounceRates
- ExitRates
- ProductRelated_Duration_Bin
- BounceRates_Bin
- VisitorType
- Month-related features
- Other engagement-related variables

These features were found to have the strongest influence on revenue prediction.

The feature selection process successfully reduced the dataset from 42 features to 10 highly informative features. This reduced computational complexity, improved model interpretability, minimized redundant information, and helped the machine learning algorithms focus on the most relevant predictors of revenue generation.

9. Conclusion

This project successfully developed a machine learning-based system to predict whether an e-commerce website visitor will generate revenue or not. The dataset was preprocessed using missing value handling, outlier analysis, encoding, feature scaling, and feature selection techniques to improve data quality and model performance.

Exploratory Data Analysis (EDA) was performed to understand customer behavior patterns and identify important relationships between variables. Feature engineering was also applied by creating custom features through binning techniques.

Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, and Gaussian Naive Bayes, were trained and evaluated. Model performance was assessed using Accuracy, Precision, Recall, F1-Score, and 5-Fold Cross Validation.

The results showed that Gaussian Naive Bayes achieved the highest Recall and Cross Validation Recall, making it the most suitable model for this project. Feature importance analysis highlighted that customer engagement-related variables played a significant role in predicting revenue generation.

Overall, the project demonstrates how machine learning can be used to support business decision-making by identifying potential revenue-generating customers. Future improvements may include using larger datasets, additional feature engineering techniques, and advanced machine learning models to further enhance prediction performance.